

Classroom Training Monitoring Tool (POE)

Date: _____

Funder: SE DFA

Trade: Electrician

Monitor: _____

Signature: _____

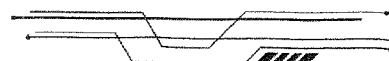
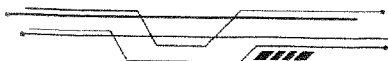
Facilitator: _____

Signature: _____

Learners: XOLANI COLLEN MALINGA

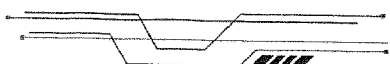
Signature: _____

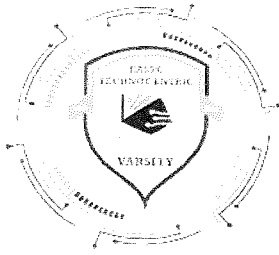
Details	Tick v (Found) X (Not Found)
Personal Details	
CV	
ID Copy	
Qualification	
SAQA/ Unit Standards	
Training Plan/ Course Layout	



Details	Learner Workbook	Summative Assessment	Performance Test Result	Practical Assessment	Appeal Procedure Form
Unit Standards/Modules:					
671101000-KM-01 Health, Safety, Quality and Legislation		✓	✓		
671101000-KM-02 Tools, Equipment and Materials		✓	✓		
671101000-KM-03: Electricity and Electronics		✓			
671101000-KM-04 Industry Context					
671101000-KM-05: Wireways and Wiring		✓	✓		
671101000-KM-06: Rotating Electrical Machinery		✓	✓		
671101000-KM-07: Electrical Supply Systems and Components		✓	✓		
671101000-KM-08: Low Voltage Protection		✓	✓		
671101000-KM-09: Fault Finding		✓			
Practical Modules					
671101000-PM-01 Use hand and power tools					
671101000-PM-02 Plan and prepare the process for the wiring, connection, testing, inspecting,					

commissioning and maintaining electrical installations and control systems					
671101000-PM-03 Prepare worksite set up for installing, wiring and connecting electrical equipment and control systems					
671101000-PM-04 Install wireways,					
671101000-PM-05 Install, wire and connect electrical equipment and control systems					
671101000-PM-06 Conduct pre-commission inspection (power on and off) fault find and test new and existing installations,					
671101000-PM-07 Carry out commissioning tests,					
671101000-PM-08 Fault find and repair electrical control systems and electrical installations					





Portfolio of Evidence

Table of Content

Tab 1: Personal Particulars

- Personal Information
- Copy of CV
- Copies of Qualifications and ID

Tab 2: Course information

- SAQA
- QCTO Implementation Plan
- Training Schedule

Tab 3: Assessments

- Formative Assessments
- Summative Assessments

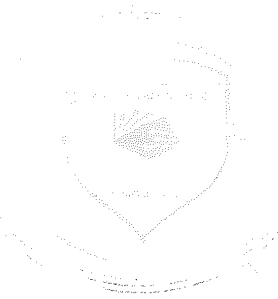
Tab 4: Practicals

- All Practical Modules

Tab 5: General

- Assessor Report

PERSONAL PARTICULARS



PERSONAL INFORMATION FORM

PERSONAL DETAILS							
SURNAME				MALINGA			
FULL NAMES				XOLANI COLLEN			
GENDER				MALE			
IDENTITY NUMBER				9511155424085			
DATE OF BIRTH (YYYY/MM/DD)				1995/11/15			
NATIONALITY				AFRICAN			
EQUITY (BLACK AFRICAN, COLOURED, WHITE, INDIAN)				BLACK AFRICAN			
HOME LANGUAGE				XITSONGA			
OTHER LANGUAGES				ENGLISH, ISIZULU			
CONTACT DETAILS							
CONTACT NUMBER				(074) 7732317			
EMAIL ADDRESS				collenx14@gmail.com			
RESIDENTIAL ADDRESS				POSTAL ADDRESS			
32467 MANDLA							
STRT EX12 ISAKANE							
TOWN		Brakpan		CODE		1550	
TOWN				CODE			
MUNICIPALITY				EKURHULENI MUNICIPALITY			
PLACE WHERE YOU BORN				GREATING			

HIGHEST QUALIFICATIONS (SCHOOL)

FULL NAME OF LAST SCHOOL ATTENDED	LUXVICUS INSTITUTE OF RRA
PLACE/ TOWN IN WHICH SCHOOL IS	Johannesburg
MUNICIPALITY IN WHICH THE SCHOOL IS	Metropolitan Municipality
HIGHEST GRADE PAST	Electrical Engineering
YEAR OBTAINED	2024
SUBJECTS PASSED	Electrical Trade Theory
	Industrial Electronics
	Engineering Science
	Mathematics

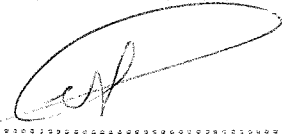
ADDITIONAL EDUCATIONAL QUALIFICATIONS

FULL NAME OF LAST EDUCATIONAL INSTITUTION ATTENDED	Artisan and Skills development Centre
PLACE/ TOWN IN WHICH LAST EDUCATIONAL INSTITUTION IS	Springs
HIGHEST QUALIFICATION OBTAINED	Boilermaker
YEAR OBTAINED	10/08/2022

NEXT OF KIN INFORMATION

RELATIONSHIP	Father
TITLE	Mr
NAME AND SURNAME	Locus Pet Malinga
CONTACT NUMBER	078 435 0026

BY SIGNING THIS, I ABOVE MENTIONED DECLARE THAT THE INFORMATION SUPPLIED IS TRUE AND CORRECT.



SIGNATURE

DATE

Building Africa one skill at a time

DATE

DATE

Curriculum Vitae of **MALINGA XOLANI COLLEN**

CONTACT DETAILS

Residential Address : 32467 Mandla Street
Extension 12
Tsakane
1550
Contact Number : 074 7732317
Email Address : collenx14@gmail.com

PERSONAL DETAILS

Date Of Birth : 15 / 11 / 1995
Marital Status : Single
Nationality : South African
Languages : English, Xitsonga & IsiZulu
Drivers Licence : Code 10 (C1)

EDUCATION

Luxvicus of Prac.Eng & Sciences
Electrical Engineering & Engineering Science
August 2024

SKILLS

- Semi Skilled Boilermaker
- Team Worker
- Adaptability
- Attention to Details

OBJECTIVE

To work in an environment which encourages me to succeed and grow professionally where I can utilize my skills and knowledge appropriately

WORK EXPERIENCE

Tsakane Super Spar
General Worker
24/01/2024 – 19/08/2024

Springs Artisan Training Centre
Semi-Skilled Boilermaker
9 Months

REFERENCES

Lehlohonolo L Mokoena
Supervisor (Super Spar Tsakane Mall)
081 481 2879

M Kgobe
Supervisor (Springs Artisan Semi-Skilled Boilermaker)
078 310 2772



REPUBLIC OF SOUTH AFRICA
NATIONAL IDENTITY CARD

Family Name
MALINGA
Given Name
KOLAM COLLEN
Sex
M
Nationality
RSA
Identification Number
9511155424095
Date of Birth
18 NOV 1995
Country of Birth
RSA
Status
CITIZEN



Conditions:

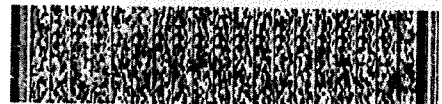
This card has been issued by the
Department of Home Affairs in terms of the
Identification Act, Act 68 of 1997

If found please return to the Department of Home Affairs
For enquiry or verification purposes contact 0800 90 11 30

Date of issue:
11 JUN 2026



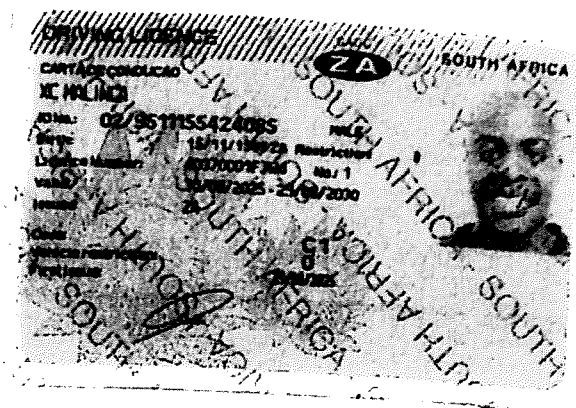
300717111



I certify that this document is a true reproduction (copy) of the original document which was handed to me for authentication. I further certify that from my observations no alteration or change was made to the original document.

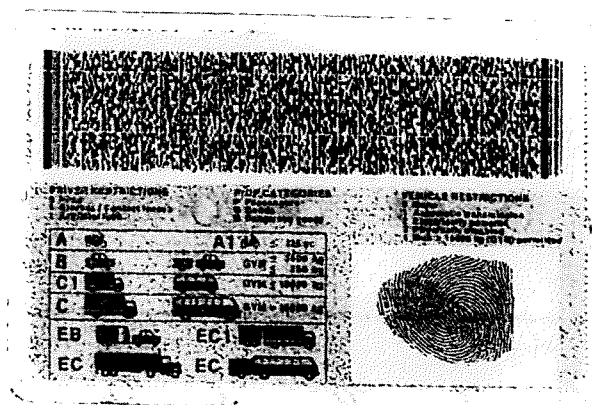
Signature.....
Force Number..... Rank.....
Name in Print.....

SOUTH AFRICAN POLICE
CLIENT SERVICE CENTRE
2026 -07- 24
TSAKAME
SUID-AFRIKAANSE POLISIEDIENS



I certify that this document is a true reproduction (copy) of the original document which was handed to me for authentication. I further certify that from my observations, no amendment or a change was made to the original document.

Signature: [Signature]
 Force Number: 02/951155424095 Rank: 1
 Name in print: Xc Mhloni



SOUTH AFRICAN POLICE SERVICE
CLIENT SERVICE CENTRE
 2026 -07- 24
 TSAKANE
SUID-AFRIKAANSE POLISIEDIENS



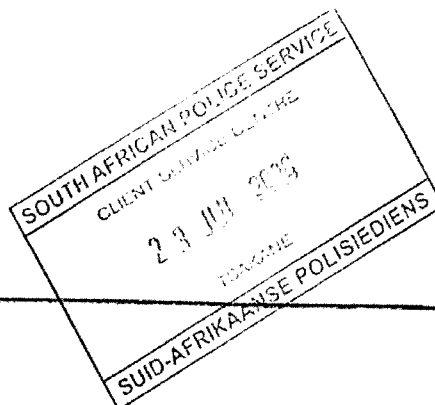
higher education
& training
Department
Higher Education and Training
REPUBLIC OF SOUTH AFRICA



EKURHULENI EAST TVET COLLEGE
"Committed to Excellence"

ARTISAN AND SKILLS
DEVELOPMENT CENTRE
10 Argon Street
Fulcrum Springs
Tel: (011) 730 6600
Fax: (011) 730 6600
E-mail: info@eec.edu.za
Private Bag X52
SPRINGS
1560

TO: TO WHOM IT MAY CONCERN
FROM: MS CN NZIMANDE
DATE: 29 July 2022
SUBJECT: CONFIRMATION LETTER



Signature: *[Signature]*
Force Number: 28170475
Name: S. B. Malaza
Rank: 6th

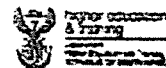
Dear Manager/Sir/Madam

This is to confirm that the student mentioned below, previously based at Artisan and Skills Development Centre, is a registered learner with Ekurhuleni East TVET College for a Boilermaker Merseta Skills Programme and has completed the following unit standards:

*Apply safety, health and environmental principles and procedures *Select, use and care for engineering hand tools *Select, use and care for engineering measuring *Select, use and care for engineering power tools *Work safely and use safety equipment *Address workplace hazards and risks *Cut materials using the oxy-fuel gas cutting process (manual cutting) *Draw and interpret simple plate pipe and structural steel plate, pipe and structural steel drawing *Measure and mark off steel profiles *Mechanically cut, drill and punch fabrication materials.

NAME: XOLANI MALINGA
ID NO: 9511155424085
PROGRAMME: Boilermaker
DURATION: 9 Months (01 November 2021-29 July 2022)
STIPEND: R1 500pm

Yours in Training and Education



2022-08-10

ARTISAN AND SKILLS DEVELOPMENT CENTRE
10 ARGON STREET, FULCRUM, SPRINGS 1560

[Signature]
Ms Cynthia Nzimande
Project Manager
Artisan & Skills Development Centre (ASDC)
Mobile: +27 73 657 0602
Office: +2711 730 6600 ext 3113
Email: cynthian@eec.edu.za

Date: 10/08/2022

I certify that this document is a true reproduction (copy) of the original document which was handed to me for authentication. I further certify that from my observations in attachment of a change was not made to the original document.

Signature: *[Signature]*
Force Number: *[Number]*
Name: *[Name]*
E-mail: *[Email]*

E-mail: info@eec.edu.za

HEAD OFFICE
Sam Ngema Road
Kwa-Thema, Springs
Tel: (011) 730 6600
Fax: (011) 730 9909
E-mail: info@eec.edu.za
Private Bag X52
SPRINGS
1560

BENONI CAMPUS
50 O'Reilly Merry Street
Northmead, Benoni
Tel: (011) 730 6600
Fax: (011) 425 3439
E-mail: info@eec.edu.za
Private Bag X004
BENONI
1500

BRACKPAN CAMPUS
58 Victoria Avenue
Brackpan
Tel: (011) 730 6600
Fax: (011) 740 9188
E-mail: info@eec.edu.za
Private Bag X10
BRACKPAN
1540

DAVEYTON CAMPUS
Hoad Street
Daveyton
Tel: (011) 730 6600
Fax: (011) 426 4091
E-mail: info@eec.edu.za
Private Bag X01
DAVEYTON
1507

KWA-THEMA CAMPUS
Sam Ngema Road
Kwa-Thema, Springs
Tel: (011) 730 6600
Fax: (011) 738 6408
E-mail: info@eec.edu.za
Private Bag X79
SPRINGS
1560

SPRINGS CAMPUS
Cnr Gold 3 Plantation Road
Springs
Tel: (011) 730 6600
Fax: (011) 362 6182
E-mail: info@eec.edu.za
Private Bag X21
SPRINGS
1560

I certify that this document is a true reproduction (copy) of the original document which was handed to me for authentication. I further certify that from my observations an amendment or a change was not made to the original document.

Signature: *[Signature]*
 Force Number: 2217104-5
 Name in print: *[Name]*



DEPARTMENT OF HIGHER EDUCATION AND TRAINING / DEPARTEMENT VAN HOËR ONDERWYS EN OPLEIDING
 REPUBLIC OF SOUTH AFRICA / REPUBLIEK VAN SUID-AFRIKA

STATEMENT OF RESULTS STAAT VAN UITSLAE

N2: ENGINEERING STUDIES (REVISED)
 N2: INGENIEURSTUDIES (HERSIEN)

AUGUST / AUGUSTUS 2024

EXAMINATION NUMBER / EKSAMENNOMMER 9511155424083

1995/11/15

XOLANI COLLEN

EXAMINATION CENTRE / EKSAMENSENTRUM
 899993877 LUXVICUS INSTITUTE OF PRAC. ENG & SCIENCE

INSTRUCTIONAL OFFERINGS / ONDERRIGAANBIEDINGE		%	RESULT/UITSLAG
11041872 ELECTRICAL TRADE THEORY N2 / ELEKTROVAKTEORIE N2	202408	53	Pass / Slaag 3
8080602 INDUSTRIAL ELECTRONICS N2 / INDUSTRIËLE ELEKTRONIKA N2	202408	69	Pass / Slaag 3
15070402 ENGINEERING SCIENCE N2 / INGENIEURSWETENSKAP N2	202408	47	Pass / Slaag 3
16030192 MATHEMATICS N2 / WISKUNDE N2	202408	55	Pass / Slaag 3

I certify that this document is a true reproduction (copy) of the original document which was handed to me for authentication. I further certify that from my observations an amendment or a change was not made to the original document.

Signature: *[Signature]*
 Force Number: *[Number]*
 Name in print: *[Name]*

SOUTH AFRICAN POLICE SERVICE
 CLIENT SERVICE CENTRE
 23 JUL 2026
 TSAKANE
 SUID-AFRIKAANSE POLISIEDIENS

*RESULT CODES APPEAR ON REVERSE SIDE / UITSLAGCODES VERSKYN OP KEERSY

DATE
 DATUM 2024/09/01

[Signature]

EXAMINATION OFFICER
 EKSAMENBEAMPTTE

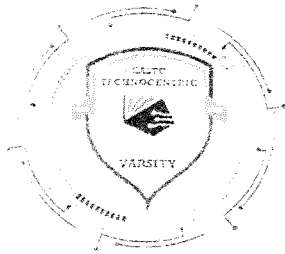
60767200M

THE DEAPRTMENT RESERVES THE RIGHT TO EFFECT CHANGES TO THIS DOCUMENT IF NECESSARY
 DIE DEPARTEMENT HOU DIE REG VOOR IN DIEN NODIG, VERANDERINGE AAN HIERDIE DOKUMENT AAN TEBRING.

Issued without alteration or erasure / Uitgereik sonder veranderings of uitwissing

11-GC03000

COURSE INFORMATION



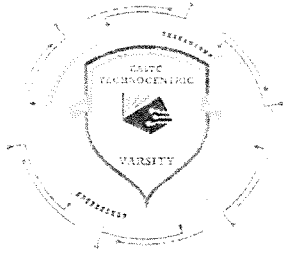
SEDFA APPRENTICESHIP 2026

ELECTRICAL APPRENTICESHIP PROGRAM COURSE LAYOUT

**SAQA ID: 91761- Occupational Certificate:Electrical
Engineering-Level 4.**

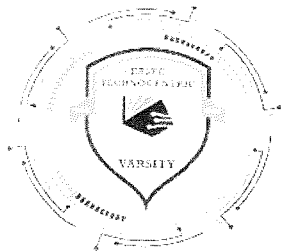
UNIT STANDARDS:

ID	UNIT STANDARD TITLE	NQF LEVEL	CREDITS
671101000- KM KM-01	HEALTH, SAFETY, QUALITY AND LEGISLATION	NQF Level 04	5
KM 671101000- KM-02	TOOLS, EQUIPMENT AND MATERIALS	NQF Level 04	8
KM 671101000- KM-03	ELECTRICITY AND ELECTRONICS	NQF Level 04	13
KM 671101000- KM-04	INDUSTRY CONTEXT	NQF Level 04	2
671101000- KM KM-05	WIREWAYS AND WIRING	NQF Level 04	11
KM 671101000- KM-06	ROTATING ELECTRICAL MACHINERY	NQF Level 04	13
KM 671101000- KM-07	ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS	NQF Level 04	31
KM 671101000- KM-08	LOW VOLTAGE PROTECTION	NQF Level 04	5



KM 671101000- KM-09	FAULT FINDING	NQF Level 04	3
PM 671101000- PM-01	USE HAND AND POWER TOOLS	NQF Level 04	22
PM 671101000- PM-02	PLAN AND PREPARE THE PROCESS FOR THE WIRING , CONNECTION , TESTING, INSPECTING, COMMISSIONING AND MAINTAINING ELECTRICAL INSTALLATIONS AND CONTROL SYSTEMS	NQF Level 04	5
PM 671101000- PM-03	PREPARE WORKSITE SET UP FOR INSTALLING WIRING AND CONNECTING ELECTRICAL EQUIPMENT AND CONTROL SYSTEMS	NQF Level 04	3
PM 671101000- PM-04	INSTALL WIREWAYS	NQF Level 04	5
PM 671101000- PM-05	INSTALL , WIRE AND CONNECT ELECTRICAL EQUIPMENT AND CONTROL SYSTEMS	NQF Level 04	38
PM 671101000- PM-06	CONDUCT PRE-COMMISION INSPECTION (POWER ON AND OFF) FAULT FIND AND TEST NEW AND EXISTING INSTALLATIONS	NQF Level 04	5
PM 671101000- PM-07	CARRY OUT COMMISSIONING TESTS	NQF Level 04	1
PM 671101000- PM-08	FAULT FIND AND REPAIR ELECTRICAL CONTROL SYSTEMS AND ELECTRICAL INSTALLATIONS	NQF Level 04	22

ASSESSMENTS

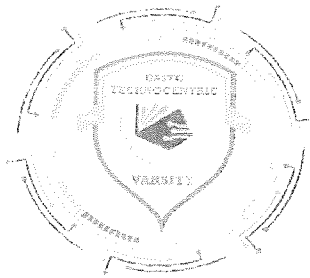


LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 01	Health, Safety, Quality and Legislation, NQF Level 4.	5



67

68

~~68~~
competent

99%

Module #	671101000-KM-01
NQF Level	level 4,
Notional hours	50
Credit(s)	Cr 5
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

HEALTH, SAFETY QUALITY AND LEGISLATION

SUMMATIVE ASSESSMENT

Details	Please Complete this Section
Name	XOLANI COLLEEN MALINGA
Contact Address	32467 MANDLA STR
Telephone (H)	0747732317
Telephone (W)	0747732317
Facsimile	N/A
Cellular	0747732317
E-mail	collenx14@gmail.com

Copyright

All rights reserved. The copyright of this document, its previous editions and any annexures thereto, is protected and expressly reserved. No part of this document may be reproduced, stored in a retrievable system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise without the prior permission.

Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

LEARNER ASSESSMENT PLAN

code	UNIT STANDARD TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <u>Xolani Malinga</u>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: Xolani Malinga

Learners Name: XOLANI COLLEEN MALINGA

Learner's signature:  Date: _____

Assessors Name: _____

Assessor's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

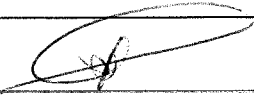
Employee witness: _____

Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <u>XOLANI COLLEEN MALINGA</u>	Signature 
Assessors Name:	Signature:

INSTRUCTIONS:

The following sections will test your knowledge, and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

LEARNING UNIT 1:

Safety, health, environment, risk and quality principles in the workplace

Learning Outcome 1:	KT0201 Legislation and regulations for workplace safety within electricity
Learning Outcome 2:	KT0202 Occupation health and safety legislation
Learning Outcome 3:	KT0203 Safety precautions and safe practices for working within industry
Learning Outcome 4:	KT0204 Personal protective equipment
Learning Outcome 5:	KT0205 Safety symbols and colour coding
Learning Outcome 6:	KT0206 Fundamentals of isolating and locking out equipment and circuits
Learning Outcome 7:	KT0207 Fundamentals of securing worksites
Learning Outcome 8:	KT0208 Protection devices
Learning Outcome 9:	KT0209 Causes, prevention and control of fires
Learning Outcome 10:	KT0210 Hazard identification and risk principles
Learning Outcome 11:	KT0211 Environmental, protection and pollution concepts
Learning Outcome 12:	KT0212 Basic first Aid
Learning Outcome 13:	KT0213 Incident reporting.
Learning Outcome 14:	KT0214 Evacuation procedures.

Task	Questions Description	Marks
1	What are the ways to reduce the risk of an electrical shock	5

Use of Circuit breakers and fuses

Proper earthing (grounding) of Electrical equipment

Wearing proper PPE such as gloves and Safety boots

Switching off and isolating power before working on equipment

Insulating live parts and covering exposed wire

11 (5)

Task	Questions Description	Marks
2	What are the responsibility for electrical installation	6

Test and Inspect the Installation before putting it into Service
 Maintain and repair any faults to ensure continued safety
 Install circuit breakers, fuse and protection devices correctly
 Use correct and approved cable, materials and protective devices
 Ensure the Installation complies with electrical safety regulations and standards

Reading and Interpreting electrical drawings/blueprints
 Calculating electrical loads to avoid overloading circuits
 Selecting correct cable sizes, circuit breakers and equipment
 Ensuring compliance with local electrical codes and regulations

6

Task	Questions Description	Marks
3	Name 5 functions of approved inspection authorities for electrical installation.	5

Inspect electrical installations conduct inspections of new and existing electrical installations to ensure compliance with regulations

Test Installations Perform electrical tests to verify safety and proper functioning according to the relevant standards e.g. SANS 10142 in South Africa.

Issue Certificates of Compliance (Coc) provide certification when an installation meets legal and safety requirements
Investigate Complaints and accidents examine electrical faults unsafe installations or incidents involving electrical systems
Report non-compliance Identify and report unsafe or illegal installations to the relevant authority

Task	Questions Description	Marks
4	Where a regulation is highlighted read it in your SABS Code of practice. Wiring techniques colors of wire-Regulation 5.2.1 What are the single-phase color code?	3

Live Phase (Brown)

Neutral (Blue)

Earth (Protective conductor) Green/Yellow

Neutral-Earth Isolation the neutral and earth must not be connected (bonded) together in a domestic installation

the green/Yellow stripe is always and exclusively used for the protective earth conductor

Task	Questions Description	Marks
5	Identify and explain 10 various types of personal protective equipment	10

Safety helmet Protects the head from falling objects and impact injuries

Safety goggles Protect eyes from dust, sparks, chemicals and flying particles.

Face Shields protects the whole from sparks, heat and chemicals

Earplugs Protects hearing from louds noise that can cause hearing loss

Respirator ~~Pro~~ Protects the lungs from dust, fumes, smoke and harmful gases

Safety boots protect feet from falling objects, sharp objects and electrical hazards

Overalls Protects the body from dirt, chemicals, heat and sparks

High Visibility Vest ~~or~~ Makes workers visble in low-light areas or near moving vehicles

Safety harness prevents falls when working at heights



Task	Questions Description	Marks
7	Give 5 safety precautions when working with electrical equipment	5

Switch ~~and~~ isolate the power supply before working on equipment

Use proper ~~personal~~ protective equipment (PPE) such as insulated gloves and safety shoes

Never work with wet hands or in wet conditions
Use insulated tools when working on electrical circuits

Inspect cables and equipment for damage before Use
Use correct tools and equipment, Use tools test for electrical work

5

Task	Questions Description	Marks
8	Name 5 unsafe conditions in a workplace	5
	Name 5 unsafe act in a workplace	5
	Name 5 points for good housekeeping	5

Poor housekeeping ✓
Slippery floors ✓
Exposed Electrical Wires ✓
Poor Ventilation ✓
damaged tools ✓

Not Wear PPE (personal protective Equipment)
Using broken or damaged tools.
horseplay ✓

Using equipment without training
 Not following safety rules and procedures

Cleaning spills immediately
 keep work areas neat and organized
 Store tools properly
 keep walkways and exits clear
 Dispose of waste regularly

(15)

Task	Questions Description	Marks
9	Write whether the statement is TRUE or FALSE	5

Protection for the human body is provided by an earthing. True if the human body is not insulated from the ground and comes in contact with the circuit, there will be a shock. True

If you do use mains voltage in a project, then you should normally use a metal case and earth case. True

A fuse contains a short piece of wire which melts if too much current flows. True

A circuit breaker contains an element that heats, bends and triggers a spring which shunts the circuits down. True

(5)

Task	Questions Description	Marks
10	Name four classes of fire and their description	4

Class A: involves solid materials such as wood, paper, cloth, rubber and plastics.

Class B: involving flammable liquids such as petrol, oil, diesel, paint and solvents.

Class C: involves flammable gases such as LPG, propane and methane.

Class D: involving combustible metals such as magnesium, aluminium and sodium.

Task	Questions Description	Marks
11	Why lockout/tag out important?	2
	What is the purpose of a lockout/tag?	2
	What does de-energization mean?	2
	What should lockout procedures and work instruction contain?	4

lockout/tag is important because it prevents accidents start-up of machinery and protects workers from injury or death while maintenance or repairs are being done.

The purpose of lockout/tag is to isolate and control hazardous energy sources so equipment cannot be switched on while work is in progress.

de-energization means disconnecting and shutting off all energy sources (electricity, gas, hydraulic) before working on equipment.

Identification of all energy sources

Steps to shut down and isolate equipment

Procedures for applying and removing locks and tags

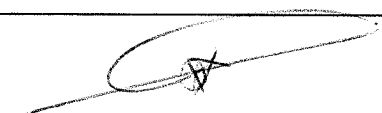
Steps to test and verify that the equipment is de-energized

110

Assessor's Report

Candidate's Name	XOLANI COLLEN MALINGA		ID No.	9511155424085
Assessor's Name			Reg. No.	
Title				
ASSESSMENT DECISION				
Outcome	C	NYC	Comments	
OVERALL COMPETENCE JUDGEMENT:				
Comments				
Date				
				
Signature of Assessor	Signature of Candidate			

Candidate Feedback Report

Candidate's Name	XOLANE COLLEEN MALINGA		ID No.	9511155424085
Assessor's Name		Reg. No.		
Title	health and Safety.			
ASSESSMENT DECISION				
Source of Evidence	C	NYC	Comment	
Knowledge questions	✓		Good	
Product /Work sample	✓		Good	
Project	✓		Good	
Indirect Evidence	✓	✱	Good.	
Overall Assessment Decision				
Additional Notes				
Date				
				
Signature of Assessor			Signature of Candidate	

Candidate Appeal Form

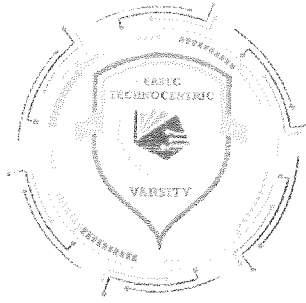
Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Title:		
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	XOLANT COLLEN MALENGA	ID No.	9511155424085
Title			
MODERATION DECISION			
Outcome	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

ASSESSMENT REVIEW

NAME of LEARNER <i>KOLANI COLLEEN MALINGA</i>		NAME of ASSESSOR	
VENUE <i>EASTC Varsity</i>		DATE of REVIEW	
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☐ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☒ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
Learner <i>SR</i>	Date	Assessor	Date
		Moderator	Date



81
Are Acceptance
Competence

FORMATIVE ASSESSMENT

HEALTH, SAFETY, QUALITY AND LEGISLATION

Module #	671101000-KM-01
NQF Level	level 4,
Notional hours	50
Credit(s)	Cr 5
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	XOLANI COLLEN MALINGA
Contact Address	32467 MANDLA STR
Telephone (H)	0747732317.
Telephone (W)	0747732317.
Facsimile	N/A
Cellular	0747732317
E-mail	collenx14@gmail.com

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions. • Ensure that all questions are completed. • Ensure that the completion of the workbook is their own work. • Ensure that all annexures are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.

Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.
-------------------	--

LEARNER ASSESSMENT PLAN

code	/MODULE TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <u>Xolani Malinga</u>	Manager:
Trainer:	Mentor / Coach:
Facilitator: <u>Balennah Tyabadi</u>	
Assessor: <u>Balennah Tyabadi</u>	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Xolani Malinga 

Learners Name: XOLANI COLLEN MALINGA

Learner's signature:  Date _____

Facilitator Name: Balennah Tyabadi

Facilitator's signature:  Date _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

Employee witness: _____

Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	☒
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	☒
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	☒
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	☒
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	☒
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	☒
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	☒
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	☒
9	Ensure the assessment environment is appropriate or make special arrangements	☒
10	Discuss moderation	☒
11	Allow the learner opportunity to clarify any items discussed	☒

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: XOLANI COLLEN MALINGA	Signature: 
Facilitator Name: Baleenah J. J. J. J. J.	Signature: 

Complete the following questions as per instructions provided

LEARNING UNIT 1:

Safety, health, environment, risk and quality principles in the workplace

Learning Outcome 1:	KT0201 Legislation and regulations for workplace safety within electricity
Learning Outcome 2:	KT0202 Occupation health and safety legislation
Learning Outcome 3:	KT0203 Safety precautions and safe practices for working within industry
Learning Outcome 4:	KT0204 Personal protective equipment
Learning Outcome 5:	KT0205 Safety symbols and colour coding
Learning Outcome 6:	KT0206 Fundamentals of isolating and locking out equipment and circuits
Learning Outcome 7:	KT0207 Fundamentals of securing worksites
Learning Outcome 8:	KT0208 Protection devices
Learning Outcome 9:	KT0209 Causes, prevention and control of fires
Learning Outcome 10:	KT0210 Hazard identification and risk principles
Learning Outcome 11:	KT0211 Environmental, protection and pollution concepts
Learning Outcome 12:	KT0212 Basic first Aid
Learning Outcome 13:	KT0213 Incident reporting.
Learning Outcome 14:	KT0214 Evacuation procedures.

Task	Questions Description	Marks
1.	Identify and explain various types of personal protective equipment	10

Hard Hat (Helmets) Protects against falling objects, impacts and electrical hazards in construction

Hearing protection: Reduces noise exposure in high-decibel

environments to prevent hearing loss

Respirators (Masks): Protects against inhaling hazardous dust, fumes, vapors or infectious agents

Safety boots: Protects against crushing injuries, slips and falling objects

Harnesses: Prevents injury from falls when working at heights

Vests/Aprons: Protects skin and clothing from hazardous substances, heat or fire

Goggles/Face Shields: Shields against flying debris and chemical splashes

Visibility Clothing: Prevent accidents by ensuring that workers are visible to vehicle operators

Task	Questions Description	Marks
2	Identify and describe Safety signs in terms of associated risk and safe conduct	10

Warning Signs - Warning you to hazards like Slippery floors or electrical dangers.

Prohibition Signs - telling you what not to do, like no smoking or no entry

Mandatory Signs: Instructing you to do something like wear PPE or follow specific route.

Information Signs providing general Safety information or directions to emergency exits

Safe Condition Show emergency exits, first aid and safety equipment location, associated risk delay response during emergency

Safe Conduct first Aid Station, emergency shower

Fire Equipment Signs showing where fire extinguishers alarm, or other safety gear are located. they help you find what you need fast in an emergency

Prohibition signs are those red circle signs with a line through them, indicating something not allowed e.g. no smoking or entry they grab attention and convey rules quickly

Task	Questions Description	Marks
3	Explain the inter-relationship between workplace safety and a productive work environment	7

Quality of Work A Safe, Organized and Clean environment

Continuous Improvement A strong safety culture encourages employees to report issues which can lead process improvements

Higher Employee Engagement Proactive safety policies demonstrate that management values their employees

Reduced disruptions and Costs A safe workplace prevents accidents and injuries

Psychological Safety and focus When employees feel secure they can focus entirely on their work rather than worrying about potential hazards

Task	Questions Description	Marks
4	Describe the regulations for the prevention and control of fires and the causes, effect and implication of fires	9

Risk Assessment are required to identify hazards and implement controls

Safety Equipment Maintenance Firefighting equipment must be installed, tested and maintained

Land Management Owners must prevent the spread of fires by maintaining firebreaks

Training and drills employees must receive fire safety training

Causes of fire

Electrical faults: Overloaded circuits, frayed

Human Negligence Improper disposal of Cigarettes unattended cooking

Industrial Hazards: dust accumulation in ventilation or sparks from machinery

Task	Questions Description	Marks
5	What are the risk of fire and explosion	8

Electrical faults or Overloads
 Open flames or Sparks ✓
 Hot Surface or
 Flammable liquids or gases
 Smoking near flammable materials
 Horseplay or Accident
 Poor Ventilation (build up of Vapours)
 Static Electricity ✓

Task	Questions Description	Marks
6	Functions of approve inspection authorities for electrical installations	12

Inspects electrical installations they examine electrical systems to ensure they comply with national regulations and safety standards

6
Test installations they conduct test to verify that installations are safe, properly wired.

Identify non-compliance they detect faults, unsafe conditions or violations of electrical regulations

Investigate incidents they may investigate electrical accidents, failures

Provide reports they prepare and submit official inspection reports to clients

Issue Certificates of Compliance After successful inspection and testing

Task	Questions Description	Marks
7	Explain and describe earthing	12

Earthing provides a safe path for fault current to flow into the ground if there is a fault

A green/yellow earth wire connects metal parts of equipment to an earth electrode

Purpose of Earthing to protect people from electric shock
Protects equipment from damage
Prevents electrical fires
Ensures proper operation of protective device

Types of Earthing

1 Rod Earthing A metal rod driven into the ground

Plate earthing ^H metal plate buried in soil

Strip/Wire earthing Metal Strip buried horizontally in the ground ✓

Connecting electrical equipment or circuits to the earth to prevent electrical shock or damage. It's like a safety net, directing stray currents harmlessly into the ground, or connecting metal parts of electrical systems to the ground.

Task	Questions Description	Marks
8	Why lockout/tag out important?	4
	What is the purpose of a lockout/tag?	3
	What does de-energization mean?	3
	What should lockout procedures and work instruction contain?	10

Why lockout/tag out it prevents the unexpected startup of machinery or release of stored energy during maintenance.

Purpose of lockout/tag to prevent injury by ensuring machines are properly shut off, to be restarted during maintenance or repairs.

De-energization means the removal of hazardous energy from machinery or equipment before lockout is applied.

lockout procedures and work instructions ✓

Method for releasing ✓

Verification procedures to ensure equipment is fully de-energized ✓

Role and responsibilities of authorized employees ✓

Steps for safely removing locks and restoring equipment to service

Steps for shutting down the equipment

Lockout procedures and Work instruction

Isolation Steps Identifying and isolation energy sources electricity, hydraulics pneumatic

Lockout/Tagout devices Using locks, tags or other devices to prevent accidental startup

Verification Steps Checking equipment is safely de-energized before work begins

Safe work procedures Outlining steps for safe maintenance or repair

Responsibilities defining who's responsible for lockout/Tagout

The Importance of Lockout/Tagout It's a procedure to ensure machines are properly shut off and can't be started accidentally while maintenance or repairs are happening. This prevents injuries and saves lives.

~~De-energization~~ De-energization means removing or disconnecting energy sources like electricity, ^{hydraulic} hydraulics or pneumatics from equipment or machinery e.g. switching everything off and making sure it stays off so you can work safely.

Task	Questions Description	Marks
9	Verify isolation explain the several ways to be followed	8

Ensure all energy sources are identified

Check that locks and tags are correctly applied

Release or control Stored energy

Test for zero energy

Use testing instruments where necessary

Visually inspect isolation points

Try-Start test

Task	Questions Description	Marks
------	-----------------------	-------

10	Name four classes of fire and their description	4
----	---	---

Class A fires these fires involve common solid combustible materials that typically leave an ash e.g wood, paper, cloth, rubber

Class B these fires are fueled by flammable liquids, combustible liquids, petroleum greases, tars

Class C fire these fires involve energized electrical equipment, wiring, circuit breakers, machinery

Class D these fires involve combustible metals such as magnesium, titanium, zirconium, sodium

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1	Legislation and regulations for workplace safety within electricity			✓
2	Safety precautions and safe practices for working within industry			✓
3	Personal protective equipment			✓
4	Fundamentals of isolating and locking out equipment and circuits			✓


Learner Signature

Date


Facilitators' Signature

Date

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONE

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes

What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method.

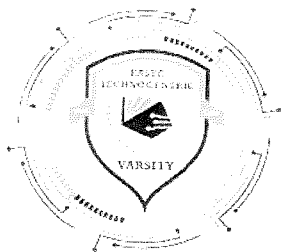
1. Written Assignment : Printed Questionnaire
2. Observation : As per Practical Assessment Guide
3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

What resources would you require to conduct the assessment?

1. Assessor toolkit
2. Theoretical assignment
3. Practical assessment
4. Learner toolkit

How would you sequence your activities? Explain

1. Record preparation process
2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment
3. Learner will write summative assessments
4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit

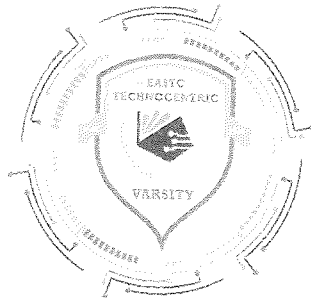


LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 2	Tools, Equipment and Materials, NQF Level 4.	8



27
30

90%

TOOLS, EQUIPMENT AND MATERIALS

FORMATIVE ASSESSMENT

Module #	671101000-KM-02
NQF Level	level 4
Notional hours	80
Credit(s)	Cr 8
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	XOLANI COLLEN MALLINGA
Contact Address	32467 Mandela Str Ex12B

Telephone (H)	0747732317
Telephone (W)	0747732317
Facsimile	N/A
Cellular	0747732317
E-mail	collenx14@gmail.com

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment
----------------	---

	criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

LEARNER ASSESSMENT PLAN

code	/MODULE TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

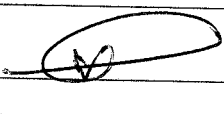
People to be involved with assessment

Learner: <u>Xolani Malinga</u>	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Xolani Malinga 

Learners Name: Xolani Malinga

Learner's signature:  _____

Date: _____

Facilitator Name: _____

Facilitator's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment

- The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
 3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
 4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

[Faint, illegible handwritten text]

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Xolani Colleen Malinga</i>	Signature 
Facilitator Name:	Signature:

Complete the following questions as per instructions provided

LEARNING UNIT 1:

KM-02-KT01: Hand tools and power tools (50%)

Learning Outcome 1:	KT0101 Types, uses and care of hand tools
Learning Outcome 2:	KT0102 Types, uses and care of portable power tools
Learning Outcome 3:	KT0103 Types, uses and care of fixed power tools

Task	Questions Description	Marks
1.	Identify portable power tools and describe their uses Identify fixed power tools and describe their uses	10

Power tools Electric hand drill Used for drilling holes in materials such as wood, materials such as wood, metal, plastic and masonry It can also drive screws.

Angle grinder Used for cutting, grinding and polishing metal, tiles and concrete often used to remove rust or smooth surfaces

Jigsaw Used for cutting curves and irregular shapes in wood, plastic or thin metal sheets

Circular Saw Used for straight cutting of wood boards and sheet materials

Heat gun produces hot air and is used for shrinking electrical insulation, loosening paint and bending plastic pipes

Fixed power tools also called stationary tools are tools that are mounted to a workbench, floor

Bench grinder Used for sharp sharpening tools, removing rust and shaping metal

Drill Press (pillar drill) Used for accurate and controlled drilling of holes in wood metal and plastic

Gable Saw Used for cutting large pieces of wood accurately and straight.

band Saw Used for cutting irregular shapes, curves and thick materials such as wood.

Belt Sander Used for smoothing and finishing surfaces especially wood and metal

Band Saw: Uses a continuous band blade to cut wood metal or plastic including curved shapes

Lathe machine Used to shape metal or wood by rotating the material while cutting tools shape

Milling machine Used for cutting, shaping and drilling metal with high precision.

10

LEARNING UNIT 2:**KM-02-KT02: Measuring and testing instruments (50%)**

Learning Outcome 1:	KT0201 Portable electrical measuring instruments
Learning Outcome 2:	KT0202 Types of electrical measuring and testing instruments
Learning Outcome 3:	KT0203 Safe use of measuring and testing instruments
Learning Outcome 4:	KT0204 Construction and operating principles of measuring and testing instruments
Learning Outcome 5:	KT0205 Methods of connecting measuring and testing instruments in circuits
Learning Outcome 6:	KT0206 Applications and methods of using electrical measuring and testing instruments
Learning Outcome 7:	KT0207 Care and maintenance of single and three phase measuring and testing instruments
Learning Outcome 8:	KT0208 Instrument transformers
Learning Outcome 9:	KT0209 Types and functions of panel mounted electrical measuring and testing instruments
Learning Outcome 10:	KT0210 Safe use of panel mounted measuring and test instruments
Learning Outcome 11:	KT0211 Protection of measuring instruments when connected in a circuit.

Task	Questions Description	Marks
2	a) Identify different types of electrical measuring instruments including fixed and portable and state the purpose. b) Explain by means of circuit diagrams how a ammeter, voltmeter is directly connected and	5

1	indirectly connected by using instrument transformers.	
---	--	--

a) Types of fixed electrical measuring instruments

1) Ammeter - Measures electrical current flowing through a circuit

Voltmeter measures electrical Voltage between two points in a circuit

Wattmeter measures electrical power consumed in a circuit

Types of Portable Electrical measuring instruments

Clamp Meter Measures current without disconnecting the circuit

Continuity tester Checks whether a circuit is Complete or broken

Multimeter Measures Voltage current and resistance in one device.

Indirect Connection: the primary winding of the Current Transformer is connected in Series with main line and Secondary winding is connected to Ammeter

Direct Connection, the ammeter is placed in Series with the circuit, and the voltmeter is connected in parallel with the load or supply

For Voltmeter the primary winding is connected parallel to the high Voltage supply and the Secondary Winding is connected to the Voltmeter

Task	Questions Description	Marks
3	Explain the function of the wattmeter (electrodynamometer type), kilowatt-hour meter, frequency meter, power factor meter and maximum demand meter	5

A wattmeter measures the electrical power in watts consumed in a circuit. It works on the electrodynamometer principle where interaction between magnetic fields of two coils produces a torque.

A kilowatt-hour meter measures the total electrical energy consumed over time. It operates on the electromagnetic induction principle. Contains Current Coil, Voltage Coil and Aluminium rotating disc.

Frequency meter measures the frequency of an alternating current supply. In a vibrating reed frequency meter, several metal reeds are tuned to different frequencies.

Power Factor Meter measures the power factor ($\cos \phi$) of an AC circuit. It uses two coils supplied with current that differ in phase.

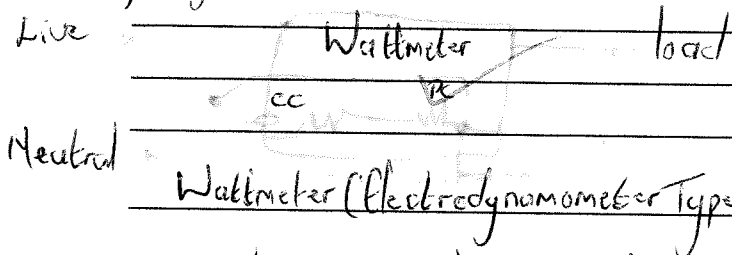
The instrument measures the phase angle between Voltage and Current. The pointer position directly indicates the power factor value.

Maximum demand Meter measures the

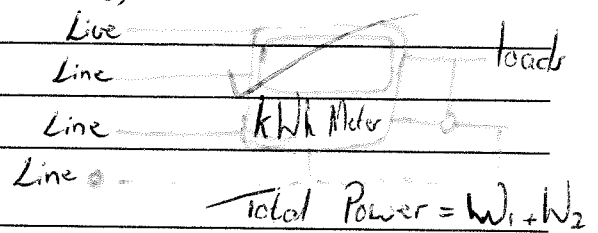
highest level of electrical demand (loads) used during a specific time period.

Task	Questions Description	Marks
4	Illustrate, by means of circuit diagrams, how the wattmeter (electrodynamometer type), kilowatt-hour meter, frequency meter, power factor meter and maximum demand meter are connected in single and three phase circuits:	10

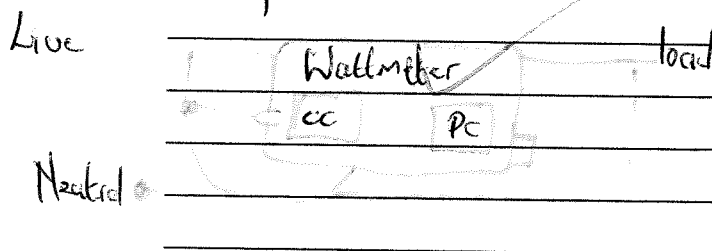
a) Single-Phase Connection



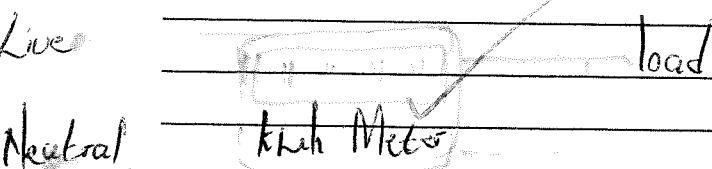
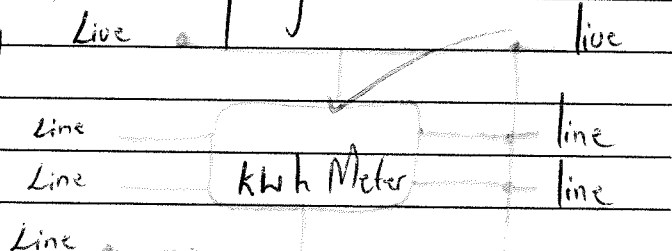
b) Kilowatt-hour (kWh) Meter



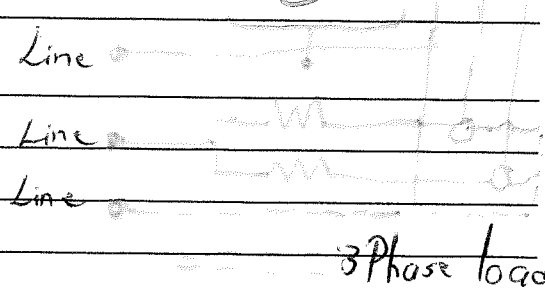
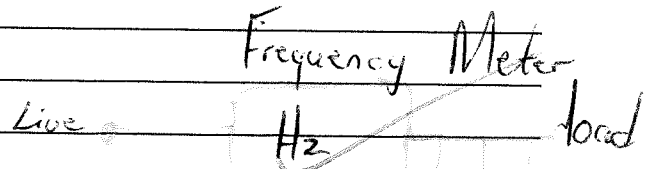
Three phase (Two Wattmeter Method)

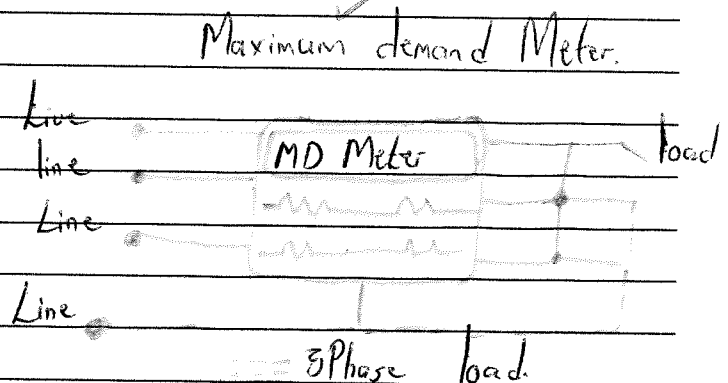
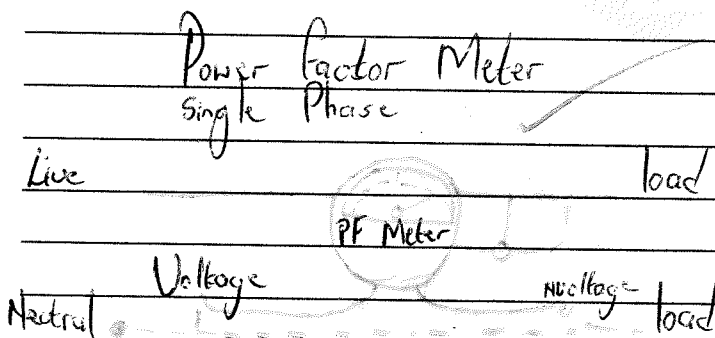
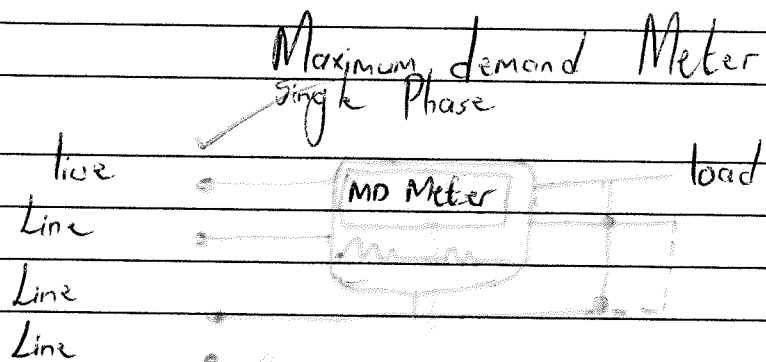
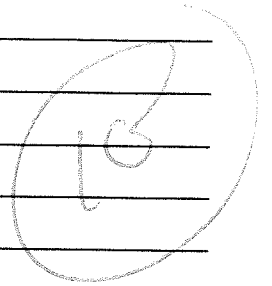
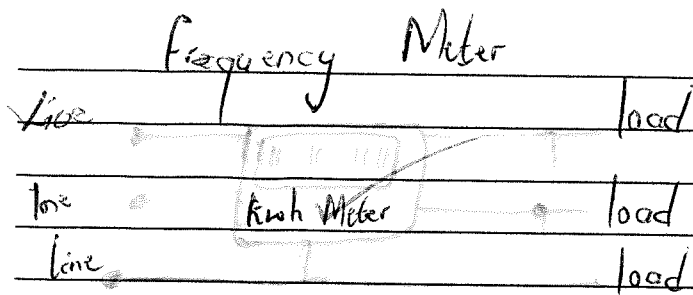


3) Frequency Meter



Single Phase (kilowatt-hour (kWh) Meter





SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1	Portable Electrical measuring Instruments			✓
2	Safe Use of measuring and testing Instruments			✓
3	Instruments transformers			✓
4	Care and Maintenance of Single-Phase			✓
	and three phase of Measuring and testing Instrum			

Learner Signature

Date

Facilitators' Signature

Date

DECLARATION BY LEARNER

I, Nelani Colten Malinga declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.

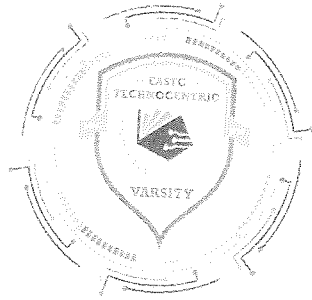
			
Learner	Date	Facilitator	Date
<u>Nelani Colten Malinga</u>	<u>10/11/2020</u>	<u>BDP</u>	<u>10/11/2020</u>

DECLARATION BY THE ASSESSOR

I, Balerinah Tjabadi (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.

			
Assessor	Date	Moderator	Date
<u>Balerinah Tjabadi</u>	<u>10/11/2020</u>		

4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit



44
145
78%

TOOLS, EQUIPMENT AND MATERIALS

SUMMATIVE ASSESMENT

Module #	671101000-KM-02
NQF Level	level 4
Notional hours	80
Credit(s)	Cr 8
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	XOLANI COLLEN MALINGA
Contact Address	32467 Mandla str
Telephone (H)	0747732317
Telephone (W)	0747732317.

Facsimile	N/A
Cellular	0797782317
E-mail	collenx14@gmail.com

Copyright

All rights reserved. The copyright of this document, its previous editions and any annexures thereto, is protected and expressly reserved. No part of this document may be reproduced, stored in a retrievable system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording or otherwise without the prior permission.

Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

6/12

LEARNER ASSESSMENT PLAN

Unit code	TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	

XM

Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

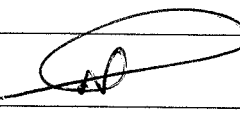
People to be involved with assessment

Learner: <i>Xolani Malinga</i>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

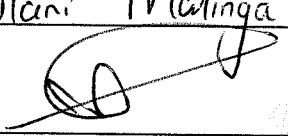
Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Xolani Malinga 

Learners Name: Xolani Malinga

Learner's signature:  _____

Date: _____

Assessors Name: _____

Assessor's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.

3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

[Faint signature]

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	☒
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	☒
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	☒
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	☒
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	☒
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	☒
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	☒
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	☒
9	Ensure the assessment environment is appropriate or make special arrangements	☒
10	Discuss moderation	☒
11	Allow the learner opportunity to clarify any items discussed	☒

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Xolani Malinga	Signature: 
Assessors Name: Bhelephahle Tshabadi	Signature: 

INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

LEARNING UNIT 1:

KM-02-KT01: Hand tools and power tools (50%)

Learning Outcome 1:	KT0101 Types, uses and care of hand tools
Learning Outcome 2:	KT0102 Types, uses and care of portable power tools
Learning Outcome 3:	KT0103 Types, uses and care of fixed power tools

Task	Questions Description	Marks
1	Identify hand tools and describe their uses	10

Hammer driving nails into wood or other materials and removing nails using the claw
breaking small objects during demolition.

Screwdriver used for tightening screws, loosening screws and assembling or disassembling equipment or furniture.

Pliers used for gripping and holding objects bending wires and cutting wires depending on the type.

Spanner/Wrench normally used for tightening nuts and bolts, loosening nuts and bolts, mechanical repairs.

Hand Saw Cutting wood, trimming timber and carpentry work.

Tape Measure used for measuring length or distance marking accurate dimensions before cutting materials.

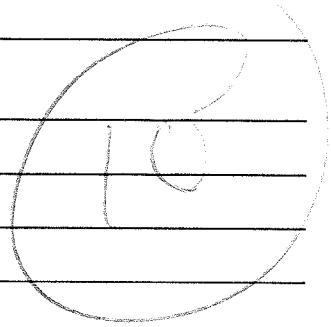
Hand tools are essential for many types of work they help with measuring, cutting, gripping, fastening and assembling materials.

Hand tools used by electricians or carpenters.

Wrench (Spanner) tightening or loosening nuts and bolts.

Allen Wrench (Hex key) driving bolts and screws
with hexagonal sockets

Try square ✓ Marking, and Checking 90° degree angles
✓ Brace and bit ✓ drilling large holes in wood using
~~man~~ manual rotation.



LEARNING UNIT 2:

KM-02-KT02: Measuring and testing instruments (50%)

Learning Outcome 1:	KT0201 Portable electrical measuring instruments
Learning Outcome 2:	KT0202 Types of electrical measuring and testing instruments
Learning Outcome 3:	KT0203 Safe use of measuring and testing instruments
Learning Outcome 4:	KT0204 Construction and operating principles of measuring and testing instruments
Learning Outcome 5:	KT0205 Methods of connecting measuring and testing instruments in circuits
Learning Outcome 6:	KT0206 Applications and methods of using electrical measuring and testing instruments
Learning Outcome 7:	KT0207 Care and maintenance of single and three phase measuring and testing instruments
Learning Outcome 8:	KT0208 Instrument transformers
Learning Outcome 9:	KT0209 Types and functions of panel mounted electrical measuring and testing instruments
Learning Outcome 10:	KT0210 Safe use of panel mounted measuring and test instruments
Learning Outcome 11:	KT0211 Protection of measuring instruments when connected in a circuit.

Task	Questions Description	Marks
2	a) Identify the basic components of various measuring and test instruments and describe the basic principle of operation. b) Describe the applications and methods of using electrical measuring and testing instruments in direct and alternating current circuits.	10

Electrical measuring and testing instruments are used to measure electrical quantities such as voltage, current, resistance, frequency and power in both direct Current (DC) and Alternating Current (AC) circuits.

Most electrical measuring instruments have similar internal parts that allow them to measure electrical values accurately.

Deflecting System This part moves the pointer on the scale when current flows.

It converts electrical energy into mechanical movement.

Controlling System This controls the movement of the pointer and brings it back to zero when there is no current.

Usually made of springs that balance the movement.

Damping System prevents the pointer from swinging too much and helps it settle quickly at the correct reading.

Indicating System Consists of pointer and Scale.

displays the measured value such as volts, amps or ohms.

Electromagnetic principle Currents flowing through a conductor produce a magnetic field.

The magnetic field causes movement of a coil or iron piece.

Electrodynamic Principle Interaction between two magnetic fields produced by coils.

Used in wattmeters to measure power.

Electrostatic Principle Uses force between charged plates.

Mostly used in high-voltage measurements.

Task	Questions Description	Marks
3	Describe safety and functionality checks to be performed on measuring and testing instruments before use	5

Inspect the body/casing for cracks, damage or loose parts
 Check the test leads and probes for cuts, exposed wires or broken insulation

Ensure the plugs and connectors are not loose/damaged.
 Make sure the display screen or scale is clear and readable.

Confirm that the fuse compartment is properly closed.
 Purpose: Prevent electrical shock, short circuits or inaccurate measurements

Turn on the instrument and check if the battery indicator shows low power

Replace the battery if the display is weak or unstable

Set the pointer to zero using the zero adjustment knob for resistance measurement, short the probes together and adjust to zero ohms

Select the correct function (Voltage, current, resistance, continuity.)

Choose the correct range for the expected measurement value.

Start with the highest range if the value is unknown

Ensure the black lead is in the COM terminal

Check continuity of the leads if necessary.

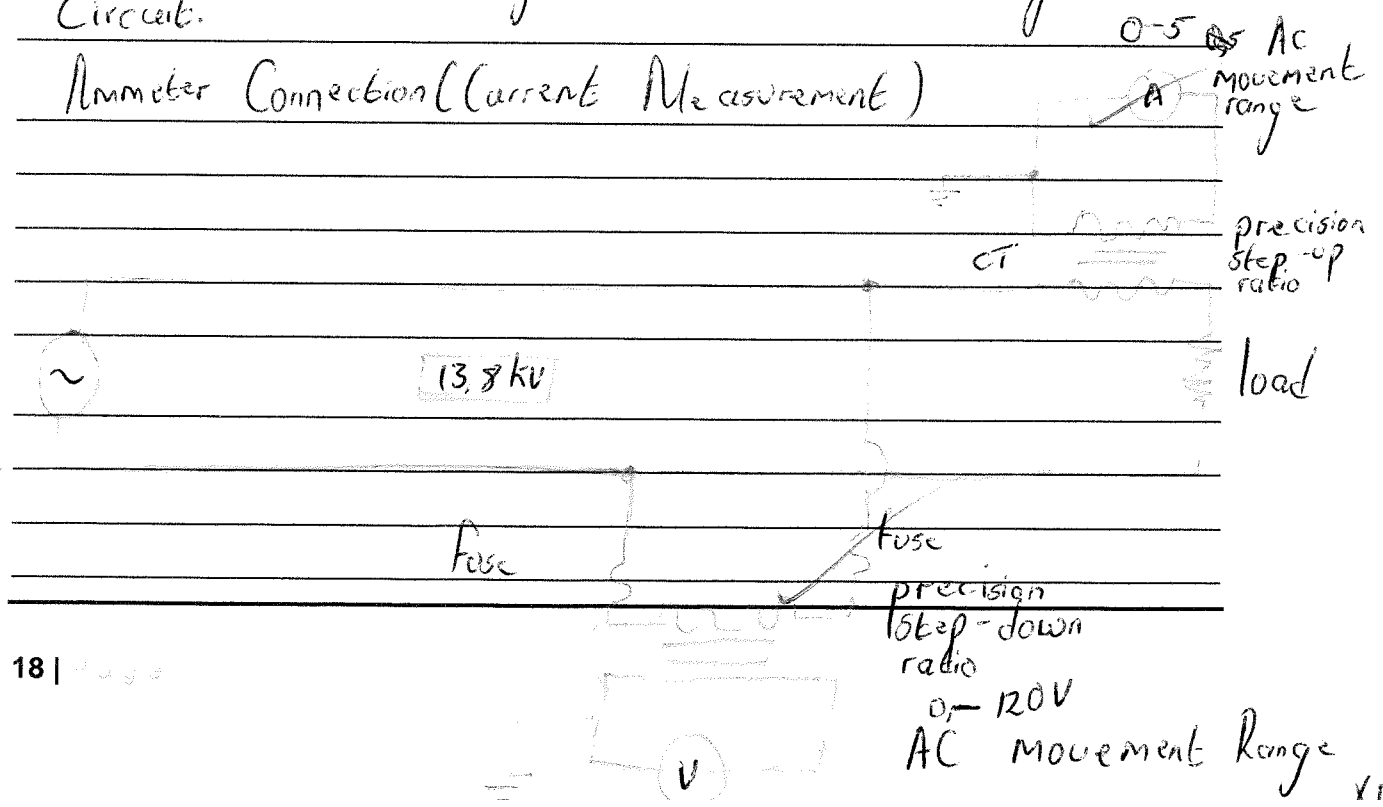
5

Task	Questions Description	Marks
4	a) Describe correct methods of handling and storing measuring and testing instruments b) Describe, with the aid of drawings, the connection of measuring instruments to single phase and three phase circuits	10

Handle instruments in their regularly care, avoiding drops shocks or vibrations
 Clean instrument regularly with appropriate materials and ensure they are dry before storage
 Store instruments in their designated cases or protective enclosures to prevent damage, dust accumulation and exposure to extreme temperature or humidity
 Calibrate instruments regularly to maintain accuracy and reliability
 Keep instruments away from magnetic fields corrosive substances and direct sunlight.

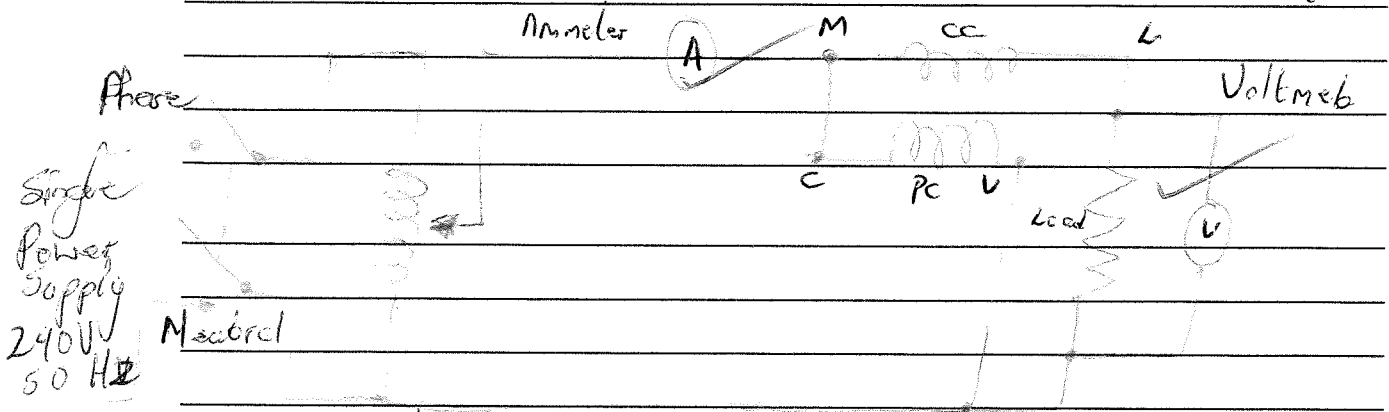
Connection of measuring Instruments in a Single-Phase Circuit.

Ammeter Connection (Current Measurement)



Description An ammeter measures electric current (I) in circuit

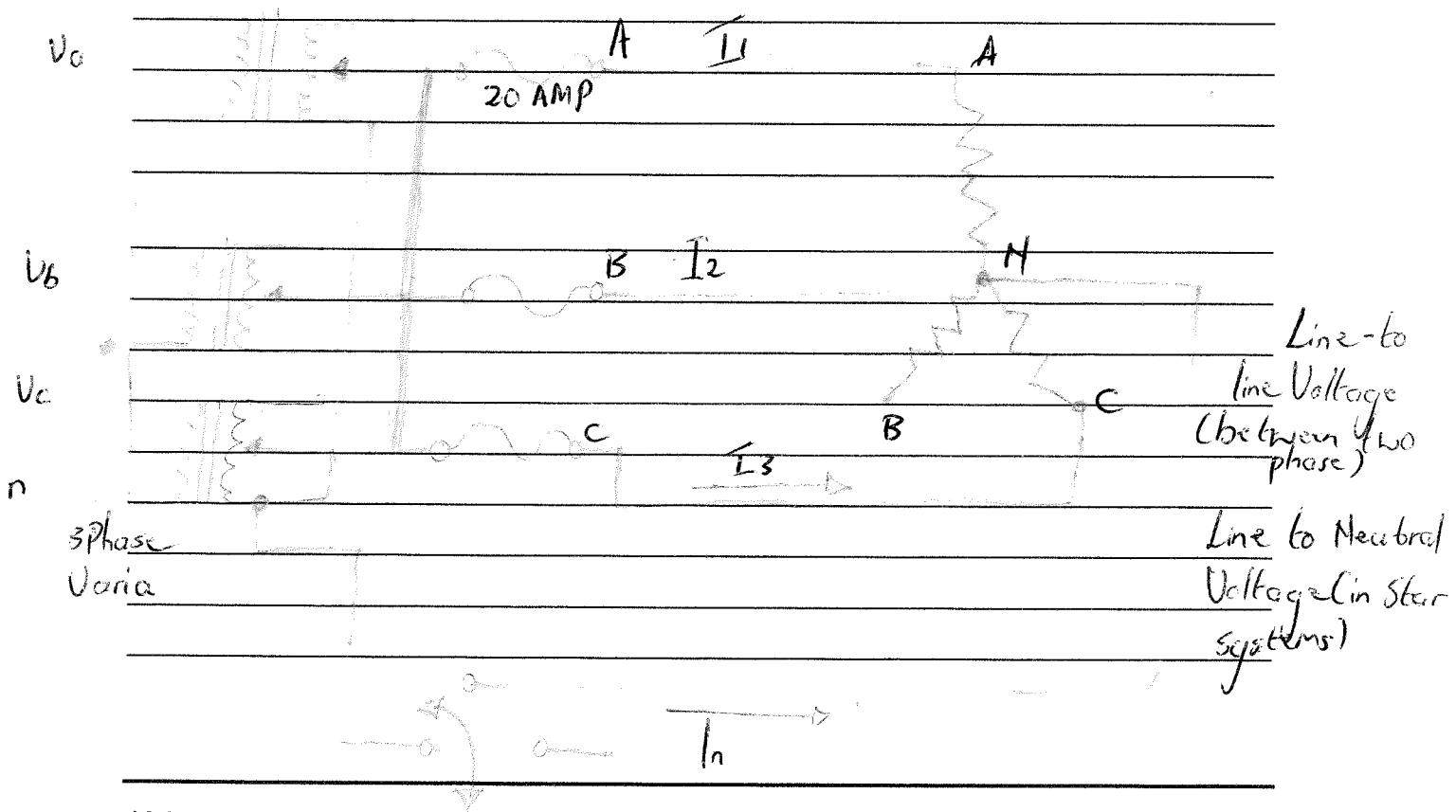
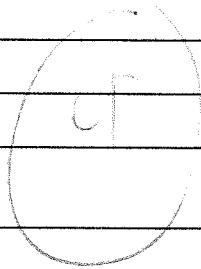
It must be connected in series with the load



Single Phase Variac

Current Coil (CC) connected in series with load

Potential coil (PC) connected in parallel



Task	Questions Description	Marks
5	Describe the various methods of protection when connecting measuring instruments in circuits	10

Use of fuses A fuse is connected in series with the measuring instrument to protect it from excessive current. If too much current flows the fuse melts and breaks the circuit preventing damage to the instrument.

Resistors are connected in series with instruments such as ammeters or voltmeters to limit the current flowing through them. This prevents overheating and damage.

A shunt resistor is connected in parallel with an ammeter. It allows most of the current to bypass the instrument so that only a small safe portion flows through the meter.

Multiplier resistors are connected in series with voltmeters to increase their voltage range and protect them from high voltages.

Before connecting a measuring instrument the correct measurement range should be selected (Usually the highest range first). This prevents overloading the instrument.

Diodes can be connected across sensitive instruments to protect them from reverse polarity or voltage spikes.

Ensuring the correct positive and negative connections prevents damage to instruments, especially analog meters.

Protective devices such as fuses, resistors, shunts, multipliers and correct operating procedures help prevent measuring instruments from being damaged by excessive current, voltage or incorrect connections.

Fuse Installed Inside Instruments protect against excessive current.

Limit Current Flow and prevent internal damage

Surge Protection device

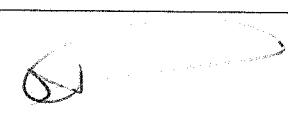
Protect Instruments from lightning Surges and Voltage Spikes.

10

Assessor's Report

[illegible]

Candidate Feedback Report

Candidate's Name	Xolani Molaga		ID No.	9511150424085
Assessor's Name	Baleenah Jabadi		Reg. No.	NAMB - A1 3025/5307
Title				
ASSESSMENT DECISION				
Source of Evidence	C	NYC	Comment	
Knowledge questions	✓		Good	
Product /Work sample	✓		Good	
Project	✓		Good	
Indirect Evidence	✓		Good	
Overall Assessment Decision				
Additional Notes				
Date				
				
Signature of Assessor			Signature of Candidate	

Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name	Bukeriyah T	Reg. No.	NAMB-A/2025/5307
Candidate's Name	Xolani Malinga	ID No.	9511153424085
Title			
MODERATION DECISION			
Outcome	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

ASSESSMENT REVIEW

NAME of LEARNER <i>XOLANI MALLINGA</i>		NAME of ASSESSOR <i>Balernah Tjabadi</i>	
VENUE <i>East</i>		DATE of REVIEW	
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
			
Learner	Assessor	Moderator	Date

